Model Comparison, Regression Prediction, and Web Application

Course: Intelligent AI and Data Engineering

Assignment Type: Group Project

Number of Groups: 8

Submission Deadline: *July 27, 2026*

Dataset links for each group ->: [Dataset Links]

Assignment Objective

The objective of this project is to apply the complete Machine Learning workflow on a real-world **regression** dataset.

Each group will:

- Understand a real-world business problem
- Perform Exploratory Data Analysis (EDA)
- Clean and preprocess the dataset
- Train and compare multiple regression algorithms
- Evaluate model performance using regression metrics
- Select the best-performing model
- Improve the model through hyperparameter tuning
- Build a web application for prediction
- Present the complete project

Project Requirements

1. Dataset Understanding

- Understand the business problem.

- Explain each feature in the dataset.
- Identify the target variable.
- Explain why the problem is a **Regression** problem.

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2. Data Preprocessing

Perform all necessary preprocessing steps.

Examples include:

- Handle missing values
- Remove duplicates
- Handle outliers (if necessary)
- Encode categorical variables
- Scale numerical features (if required)
- Feature engineering (optional)
- Split data into training and testing sets

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3. Exploratory Data Analysis (EDA)

Include appropriate visualizations such as:

- Histograms
- Boxplots
- Scatter plots
- Correlation Heatmap
- Distribution Plots
- Pair Plots (optional)

Explain the important insights discovered from the data.

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4. Regression Models

Each group must compare **at least 7 regression algorithms**.

Required algorithms:

- Linear Regression
- Ridge Regression
- Lasso Regression
- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting Regressor
- Support Vector Regressor (SVR)

Optional (Bonus):

- XGBoost Regressor
- Extra Trees Regressor
- AdaBoost Regressor
- CatBoost Regressor
- LightGBM Regressor

5. Model Evaluation

Evaluate every regression model using:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Explain:

- What each metric measures
- Which values indicate better performance
- Why the selected model performed best

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6. Regression Model Comparison

Create comparison tables and charts including:

- Algorithm Name

- Training Time
- MAE
- MSE
- RMSE
- R^2 Score
- Advantages
- Disadvantages

Finally, justify why your selected model is the best choice for the dataset.

Web Application

Each group must build a web application that allows users to make predictions using the best regression model.

The application should include:

- Home Page
- Project Description
- Dataset Information
- Model Information
- Prediction Form
- Prediction Result
- Model Comparison Page
- About Team Page

Technical Requirements

The project must include:

- Clean and organized code
- Proper comments
- requirements.txt
- README.md

- GitHub Repository
- Saved trained model
- Deployment link

Deliverables

1. GitHub Repository

Include:

- Source Code
- Dataset (or dataset link)
- README.md
- requirements.txt

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2. Technical Report (5–10 Pages)

The report should include:

- Introduction
- Problem Statement
- Dataset Description
- Data Cleaning
- Exploratory Data Analysis
- Feature Engineering
- Regression Algorithms
- Hyperparameter Tuning
- Model Evaluation
- Model Comparison
- Best Model Selection
- Deployment
- Conclusion
- Future Improvements

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3. Presentation Slides

Maximum:

15–20 slides

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4. Live Web Application

Live demonstration of the prediction application.

Group Dataset Allocation

Group	Dataset	Target Variable
Group 1	House Prices	Sale Price
Group 2	California Housing	Median House Value
Group 3	Medical Insurance Cost	Insurance Charges
Group 4	Student Performance	Final Grade
Group 5	Bike Sharing Demand	Rental Count
Group 6	Used Car Price Prediction	Selling Price
Group 7	Energy Efficiency	Heating Load
Group 8	Diamonds Price Prediction	Price

Presentation Requirements

Each group has **15–20 minutes**.

Present the following:

1. Business Problem
2. Dataset Overview

3. Data Cleaning
4. Exploratory Data Analysis
5. Feature Engineering
6. Regression Algorithms Used
7. Hyperparameter Tuning
8. Model Evaluation
9. Model Comparison
10. Best Model Selection
11. Live Prediction Demo
12. Challenges Faced
13. Future Improvements

Every group member must participate.

Evaluation Rubric (100 Marks)

Criteria	Marks
Problem Understanding	10
Data Cleaning & Preprocessing	10
Exploratory Data Analysis	10
Feature Engineering	10
Regression Model Development	15
Comparison of 7+ Regression Algorithms	20
Model Evaluation & Interpretation	10
Web Application	10
Presentation	5

Bonus Marks (+10)

Bonus marks may be awarded for implementing additional features such as:

- SHAP Explainability
- Feature Importance Visualization
- Cross Validation
- Learning Curves
- Residual Error Analysis
- Docker Deployment
- User Authentication
- Database Integration
- Prediction History
- Attractive UI/UX

Learning Outcomes

Upon successful completion of this project, students will be able to:

- Understand real-world regression problems.
- Perform complete data preprocessing.
- Compare multiple regression algorithms.
- Evaluate regression models using appropriate metrics.
- Improve model performance through hyperparameter tuning.
- Select the best-performing regression model.
- Build and deploy an end-to-end regression prediction system.
- Present technical work professionally.
- Apply Machine Learning concepts to solve real-world prediction problems.

Submission Checklist

Before submission, ensure you have included:

- GitHub Repository
- Dataset

- Source Code
- requirements.txt
- README.md
- Technical Report
- Presentation Slides
- Trained Model (.pkl/.joblib)
- Regression Model Comparison Results
- Working Prediction Web Application

Good luck! Remember: the goal is not only to achieve the lowest prediction error but also to understand *why* one regression model performs better than another and to demonstrate the complete Machine Learning workflow from data preparation to deployment.